

DLAV: Homework 2 Report

Part I: Softmax Exercise Results and Parameter Tuning

In this section of the code, we implemented a softmax classifier trained on the CIFAR10 dataset. The goal was to classify input images into 10 different classes. To prepare the data, we loaded the CIFAR-10 dataset, which consists of 60,000 32x32 color images in 10 classes, with 6,000 images per class, and split the dataset into training and validation sets.

We then implemented a softmax loss function that computes the loss and gradient for the softmax classifier. Using stochastic gradient descent (SGD), we trained the softmax classifier by optimizing the loss function. We experimented with different learning rates and trained the model for 2500 iterations.

We used the validation set to tune the learning rate hyperparameter, testing two different learning rates: $1e-7$ and $5e-4$. For each learning rate, we trained the softmax classifier and evaluated its performance on both the training and validation sets. After training, the results showed that a learning rate of $5e-4$ achieved the highest validation accuracy of 40.5%.

Part II: Linear Classification Exercise Results and Parameter Tuning

In this part of the code, we implemented linear classification using PyTorch. The neural network consists of an input layer with 3072 features, two hidden layers each containing 300 neurons, and an output layer with 30 classes. ReLU activation functions were applied between the layers to introduce non-linearity. We trained the neural network using stochastic gradient descent, iterating through the training dataset in batches, computing the forward pass, loss, and performing backward pass and weight updates.

During training, we observed the performance of the model on the validation set and adjusted the hyperparameters accordingly, landing on a learning rate of $8e-3$. The training process yielded an accuracy of around 50-53% by the end of it.

Part III: Evaluation and Comparison of Models

Two models were evaluated using the CIFAR-10 dataset: one implemented in PyTorch and another using softmax regression.

The PyTorch model consisted of two hidden layers with ReLU activation functions, and used a trained PyTorch model to predict classes for input data. It achieved an accuracy of 50.65% on the test set.

The Softmax Regression Model implemented softmax regression with cross-entropy loss, and used the learned weights to compute class predictions. It attained an accuracy of 39.75% on the test set. The Softmax regression, while simpler, demonstrated lower accuracy, indicating its limitations compared to deeper models like the PyTorch model.